01 / PROBLEM · 00-01 MIN · FICTIONAL PROSPECT

Know when to inspect.
Know when not to guess.

NorthSea Integrity Operations monitors subsea gas pipelines. Its engineers need to distinguish a possible leak from unreliable sensors before mobilizing an offshore investigation.

Possible release?

Corroborate physical evidence across usable sensors and failure domains.

Where to inspect?

Offer a compatible pipeline candidate only when the numerical fit supports it.

Can we trust the data?

Expose gaps, conflicts and competing explanations. Unknown is not healthy.

Synthetic data only | Read-only decision support | Human safety authority

02 / PROBLEM · 01-03 MIN

The prospect is buying less investigation risk.

False alarm

Unnecessary offshore personnel, vessel or ROV mobilization.

High inspection cost, operational disruption and declining trust in alarms.

Missed release

Environmental harm and liability; oil-line spills add cleanup exposure.

Serious regulatory violations can threaten licences, including possible revocation under applicable law.

> "Our dashboard is cheaper. Show that AI reduces triage effort without hiding real leaks or exporting our operational data."

Do not optimize alarm reduction while **concealing missed-event risk.**

Mobilization economics are discovery assumptions. Licence revocation is conditional, not automatic; source boundaries: engineering report.

03 / PROBLEM · 03-05 MIN

One decision. Two acceptance contracts.

Head of Subsea Operations

Budget owner: inspection effort, adoption, delivery time and accountability.

"What will four weeks prove, and who owns a wrong recommendation?"

Integrity & Instrumentation Lead

Calibration, provenance, failure domains, environmental confounders and reproducibility.

"Could a shared gateway explain these alarms? Show me the raw evidence."

My RE 2023 research motivates **accuracy, currentness, completeness and consistency.** The demo's localization is a new simplified extension, not a thesis-validated capability.

Thesis Paper C, PDF pp. 162-164; defense slides 16, 18-21 and 47. Prospect and requirements are hypotheses, not a claimed customer interview.

04 / LIVE DEMO · 05-07 MIN

Watch the window move.
Then pin the answer to evidence.

01 / Replay

24 nodes; acoustic power, methane and temperature. LIVE mode; accelerated synthetic time.

→

02 / Inspect

60-second window, 5-second hop. Pause; select a sensor; show all three charts and raw records.

→

03 / Investigate

Ask a question. Show its pinned window, role tasks, checked answer and actual model receipts.

Numerical monitoring keeps running **independently of hosted inference.**

Open live console Retained walkthrough evidence

Live answers can take tens of seconds. If the provider is unavailable, show retained evidence explicitly as recorded; never relabel SCRIPTED as LIVE.

05 / LIVE DEMO · 07-11 MIN · S1

Corroboration supports a suspected leak.

Ask the operator's questions

1. "The alarms started together. Is this a possible leak or an instrumentation issue?"
2. "Which pipeline section should our integrity engineer inspect first, and why?"

Open Agents: quality evidence, diagnosis dependency, accepted response, pinned scope and real CLI model identity.

Retained LIVE snapshot / S1 / 00:03:00

Suspected leak, not confirmed.
P1: compatible candidate at **375m**.
Fit residual: **0.162**.
Usable coverage: **99.3%** (4289/4320).

Not a calibrated confidence interval. Only 8/72 channel-health claims reviewed in this bounded answer; the full report is retained.

The LLM did not calculate 375m. **The numerical fit did.**

Values above are from a retained pinned window, not guaranteed values for every live window: visible-e2e.json.

06 / LIVE DEMO · 11-15 MIN · S2 + S5

Scattered faults are different.
Missing coverage is not an all-clear.

S2 / Instrumentation issue

"Which sensors are unreliable, and what evidence is missing?"

Retained coverage: **95.3%**. Quality facts are supported; no supported inspection location.

Inspect scattered gaps/spikes and competing physical evidence.

S5 / Too few observers

"Can you locate a leak with this missing data?"

Retained coverage: **8.3%**. Physical-event/location abstention; observed quality facts remain reportable.

A physical signal with too few usable observers is not enough to diagnose or locate.

An accepted abstention is **a useful, checked answer** — not a successful leak diagnosis.

Retained S2/S5 scopes: visible-e2e.json; final answer outcomes: agents-live.json.

Two execution loops.
One immutable evidence contract.

Source + numerical loop	Synthetic arrivals → canonical buffer → 60s/5s/5s windows → quality, hypothesis and fit ledger.
Durable boundary	PostgreSQL 16: versioned snapshots, scoped task state, receipts and LangGraph checkpoints.
Agent worker	Response workplan → quality → diagnosis → checked response. LangChain wraps hosted Copilot CLI inference.
Operator + telemetry	FastAPI/Pydantic → authenticated SSE → TypeScript/esbuild console. Native OTel → local Langfuse/Jaeger.

CustomerOps-style durable authority. **NumPy adds the marine numerical layer.**

Bad data stays visible.
It does not become leak evidence.

Canonicalization contract

- Known units and explicit timezones normalize; unknown formats are not guessed.
- Keep raw provenance. Flag gaps, spikes, late arrivals and duplicate conflicts.
- Conflicting values cannot silently corroborate a release.

Event-time contract

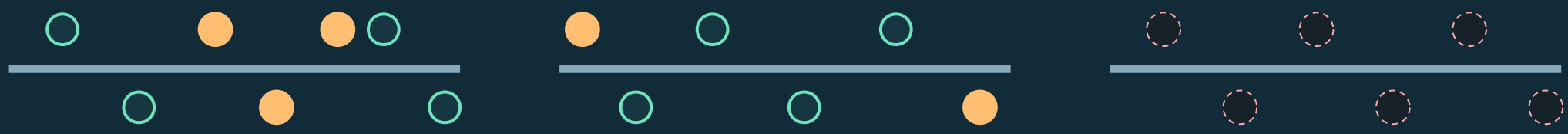
```
Window: [end - 60s, end)
Hop: 5s   Allowed lateness: 5s
Use only observations released
by the source clock.
```

Late evidence creates immutable revisions. Baseline pre-roll must be verified normal and available before the replay epoch.

Quality is **use-case-specific**, not a single "good/bad sensor" score.

ingest.py diagnostics.py monitoring.py

Simultaneous anomalies are not sufficient.



Qualitative illustration, not copied defense geometry or a calibrated spatial model.

```
z = (value - baseline median) / scale
scale = max(1.4826 × MAD, channel floor)
```

Persistent usable acoustic excursions + methane corroboration + neighborhood support + independent failure domains. Check counterevidence before allowing a physical hypothesis.

Missingness alone never proves a leak. **A shared failure can also be spatially coherent.**

A reproducible inspection candidate. Not a confidence interval.

```
k_i(x) = 1 / (distance_i(x)^2 + r0^2)
w_i = quality weight_i / scale_i^2
A = max(0, sum(w_i k_i y_i)
             / sum(w_i k_i^2))
residual = sqrt(
  sum(w_i (y_i - A k_i)^2) / n)
```

Inspectable search + rejection

- Search relevant centerlines on a 25m grid; fit nonnegative source strength.
- Withhold for insufficient usable observers or poor fit.
- Retain compatible candidates; expose multiple-segment ambiguity or excessive width.

Currents, mixing, attenuation, calibration and geometry need field validation. Grid spacing is not accuracy.

Matched synthetic generation explains ideal zero error. **It does not prove field localization.**

localization.py: locate() synthetic.py: assumptions

Three scoped specialists.
One checked operator answer.

Sensor Quality

Coverage, conflicts, sensor patterns and neighbors.

Can support observed quality facts without proving a physical event.

Event Diagnosis

Consumes accepted quality; inspects hypothesis, counterevidence and numerical fit.

Cannot turn missingness into a categorical leak/location.

Operator Response

Owns the workplan, routes/reuses compatible specialist results and synthesizes.

Returns accepted claim IDs and allowed review recommendations.

Workplan → quality → diagnosis → synthesis. **Real abstention propagates.**

Application-owned, audited mandatory intake; separate live role interpretation. Deterministic checks verify scope, claim kind, evidence membership and certainty. Readable text is rendered from accepted claims.

role authority workflow claim checks

Execution must survive interruption.
The evidence must survive review.

Bounded, durable work

- LangGraph interrupt/checkpoint before external I/O; durable task/receipt transactions.
- Generation fences and idempotent intake IDs bound replay and stale workers.
- Root: at most 2 children, 10 model calls, 18 tool calls, 300 seconds.

Observable failure

- Auth/quota block is explicit; numerical monitoring continues.
- OTel spans show agent, generation, tool and guardrail in Langfuse/Jaeger.
- Stored observations are verified; exporter HTTP success alone is insufficient.

An evidence layer needs **recoverable state and truthful counters**, not just agent prompts.

Local Langfuse workflow.py telemetry.py Trace appendix

Evidence of a working demo.
Not evidence of field reliability.

27/27
Final LIVE accepted-correct

9 scenario/question cases, repeated 3 times. Includes required abstentions. Actual gpt-6.1-sol receipts.

120
Numerical mature windows

20 seeds × 6 scenarios. Controlled leak precision/recall 1.0/1.0; not operational probabilities.

0.8 / 1.0
Fault precision / recall

100 false-positive fault attributions retained. Ideal zero location error reflects matched assumptions.

167 Python regressions; 4 browser cases; Ruff, strict mypy, TypeScript/build. Mismatch stress: 18 locations, 2 abstentions, p95 error 25m. Compute p95 0.476s is not detection latency.

The early LIVE cohorts failed **2/27 → 24/27 → 27/27**. Failures are retained, not overwritten.

Full results and boundaries · failed cohort · final cohort

Use the LLM where language helps. Keep physics and authority outside it.

Boundary	Choice	Reason / cost
Continuous detection + fit	Robust statistics, rules, numerical search	Reproducible, fast, inspectable; simplified physics needs validation.
Question interpretation + role work	Hosted Copilot CLI, gpt-6.1-sol	Flexible language; latency, allowance and privacy constraints.
Accepted claim + permissions	Typed schemas, deterministic checker, static registry	The model cannot invent accepted metres or extend authority.

LIVE roots with fresh specialists: **4.27 calls / 60.49s mean**; compatible reuse: **2 calls / 29.21s**. Same-evidence single-agent baseline: **1 call / 19.13s**.

Multi-agent buys **role isolation and durable audit**, not demonstrated speed or accuracy improvement.

Different fresh/reused cohorts, not a controlled latency experiment. Provider USD cost is unknown: RESULTS.md.

Measure triage value.
Do not sell imagined avoided spills.

Illustrative capacity calculation

```
120 alerts/month × 0.25h saved/alert
× USD100/h × 12 months
= USD36,000/year
```

All inputs are assumptions. Capacity released is not automatically cash saved; inspection avoidance is not guaranteed.

Customer numbers needed

- Reviewed alert volume and median investigation effort.
- Mobilization cost, escalation workflow and analyst adoption.
- Integration/support effort, hosted allowance and deployment constraints.

Compare against the existing workflow. **Never trade safety sensitivity for a prettier ROI.**

A four-week shadow pilot.
One network. Explicit stop gates.

Week 1

Connector/schema, baseline, failure-domain map and security review.

Gate: approved data path and usable source contract.

Week 2

Historical replay; expert adjudication; known gaps and confounders.

Gate: labels/tests support separate error analysis.

Week 3

Read-only shadow workflow; operator review and escalation ownership.

Gate: evidence is usable and every action stays human-approved.

Week 4

Joint baseline comparison; triage, false alarms, sensitivity, coverage and latency.

Gate: continue, revise or stop.

Commit only after **scope, pricing, safety thresholds and acceptance criteria** are jointly agreed.

Conditional plan, not a guaranteed integration deadline. No sensitivity claim without labeled events or validated tests.

Keep the safety workflow.
Add a reviewable evidence layer.

What is true today

Original synthetic demo, scoped read-only tools, authenticated console, durable evidence and hosted Copilot inference.

No historian/platform connector, certified leak protection, control endpoints, production identity or security certification.

What must be approved

Customer deployment boundary, source connector, data retention and inference destination.

This demo is **not local-only**. If hosted inference is unacceptable, resolve the architecture before real data is used.

> "Will you sponsor a scoped shadow pilot, subject to integrity and security approval, with agreed stop gates and a joint baseline review?"

Expert knowledge becomes **reviewable criteria and evidence** — not unchecked model authority.

18 / OUT OF ROLE · 55-60 MIN

Questions for AITOMATIC.

1. How do your strongest FDEs turn an expert's operating judgment into governed agent behavior — and who owns ongoing validation?
2. Where do customer pilots most often stall: data access, integration, evidence quality or workflow adoption?
3. What would you expect me to own from discovery through production handoff in my first 90 days?

5m	10m	30m	10m	5m
Problem 00-05	Demo 05-15	Whiteboard 15-45	Business 45-55	Questions 55-60

Private repository One-page brief Technical appendix

APPENDIX A / ON DEMAND · NO EXTRA SCHEDULED TIME

Test the counterexamples.
Not just the easy leak.

Scenario	Challenge	What the reviewer should inspect
S1	Coherent leak-compatible evidence	Usable acoustic/methane support, compatible position, uncertainty.
S2	Scattered sensor faults	Quality patterns and unsupported physical/location claims.
S3	Shared gateway / full-source outage	Advancing source clock despite missing arrivals; physical state unknown.
S4	Broad environmental disturbance	Counterevidence and missing methane corroboration.
S5	Physical signal with too few observers	Coverage threshold and physical/location abstention.
S6	Verified-normal variability	Normal baseline behavior without fabricating a physical event.

Six scenarios are synthetic. Final LIVE question qualification covers S1/S2/S5; the numerical hold-out covers all six. These are different denominators.

APPENDIX B / ON DEMAND

Can you prove the agents actually ran?

Receipts, not role labels

- LIVE mode, requested/observed gpt-6.1-sol, identity source cli_event.
- Task/generation/window scope, model/tool counters and durable accepted result.
- Reuse linked explicitly; not counted as a new specialist model execution.

Stored observability

Verified final cohort trace:

> `f00e9c77773918c0fbaa9b7f0e35fa2d`

Agent, generation, tool and guardrail observations in Langfuse. Fresh browser audit feeds start at the current cursor; historical backend events remain separately retrievable.

Visible walkthrough evidence was **recovered across sessions**, not an uninterrupted six-question run.

Final LIVE cohort Visible walkthrough Qualification boundaries

APPENDIX C / ON DEMAND

Failures changed the implementation. They did not change the truth standard.

Observed failure	Root-cause correction	Evidence boundary
2/27 initial LIVE answers	Audited application intake, exact schemas and final-turn protection.	Failed receipt retained; tool/model budgets not inflated.
24/27 intermediate cohort	Quality facts separated from physical certainty; real abstention still propagates.	No relaxed physical-event/location gate.
Missing checkpoint bootstrap	Explicit migration, fail-fast startup, fresh/reopen/handoff tests.	Integrated database proof beyond isolated unit tests.
Long ingest lock / UI race	Bounded idempotent chunks; Agents focus before async task read.	Targeted contention and held-read browser regression.

Senior FDE ownership means **reproducing, fixing and retaining evidence**, not hiding demo failures.

APPENDIX D / ON DEMAND

Implemented boundaries. Unfinished production work.

Implemented in the demo

- Argon2 login, scoped cookies/CSRF and authenticated API/SSE.
- Forced tenant/site PostgreSQL RLS; separate tool/checkpoint roles.
- Static read-only authority; inference subprocess strips app/database/viewer credentials.
- Loopback-bound local services and secrets outside the repository.

Not certified or integrated

- Production SSO/multi-user identity, HA and security certification.
- Historian, Falkor or DanaOS connector.
- Field-calibrated gas/oil detection or localization.
- Local-only inference or autonomous shutdown/control.

Research grounding: thesis Paper C (RE 2023), defense 16/18-21/47. Public industry positioning: Falkor. Regulatory scenario: archived Norwegian Petroleum Act translation, sections 10-13/10-16; not current legal advice.

Original deck and diagrams. CustomerOps informs customer-first narrative, scoped specialists, durable evidence, failure ownership and explicit tradeoffs; its content/assets and private discussions are not copied. Detailed source boundaries: REPORT.md.